EARLY WARNING SYSTEM FOR ACUTE FOOD CRISIS

Using Climate Shocks, Market Prices & Conflict Data

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Date: Feb 28, 2026

PROBLEM STATEMENT

- Kenya faces recurring droughts, high food prices, and localized conflict.
- Food security reports are mostly descriptive (reactive).
- No integrated predictive model combining climate, market, and conflict indicators.
- Need for a data-driven early warning system.

OBJECTIVE OF THE PROJECT

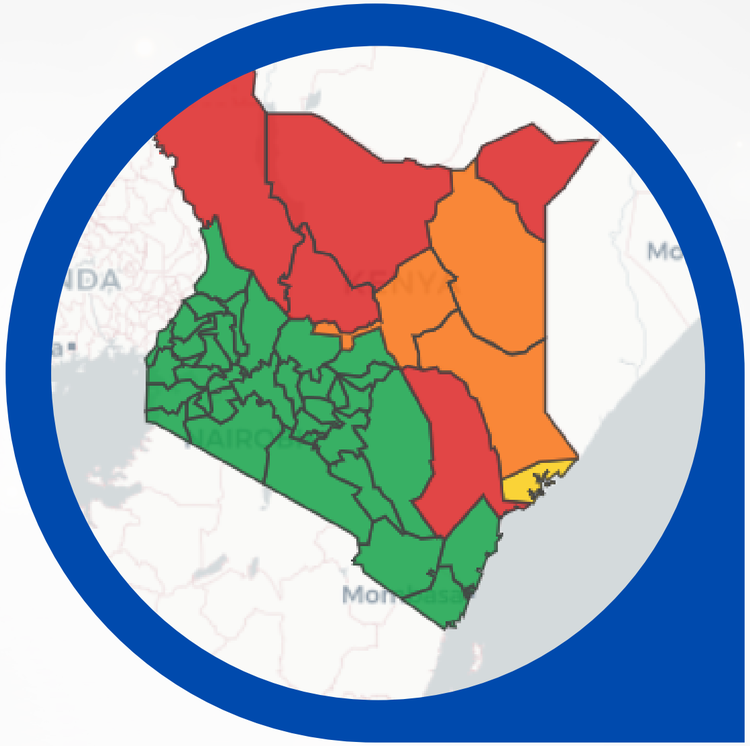
Main Goal:

Predict probability of a county entering IPC Phase 3+ (Crisis or worse) one month ahead.

Specific Objectives:

- Build a county-month dataset
- Engineer lagged climate & market features
- Compare ML classification models
- Identify key drivers of food insecurity

DATA SOURCES



Target Variable

- Integrated Food Security Phase Classification (IPC Phases)

Predictor Variables

- Climate: Rainfall anomalies, NDVI
- Market: Monthly maize retail prices
- Conflict: Event counts & fatalities
- Panel structured as county-month dataset

Risk Bands

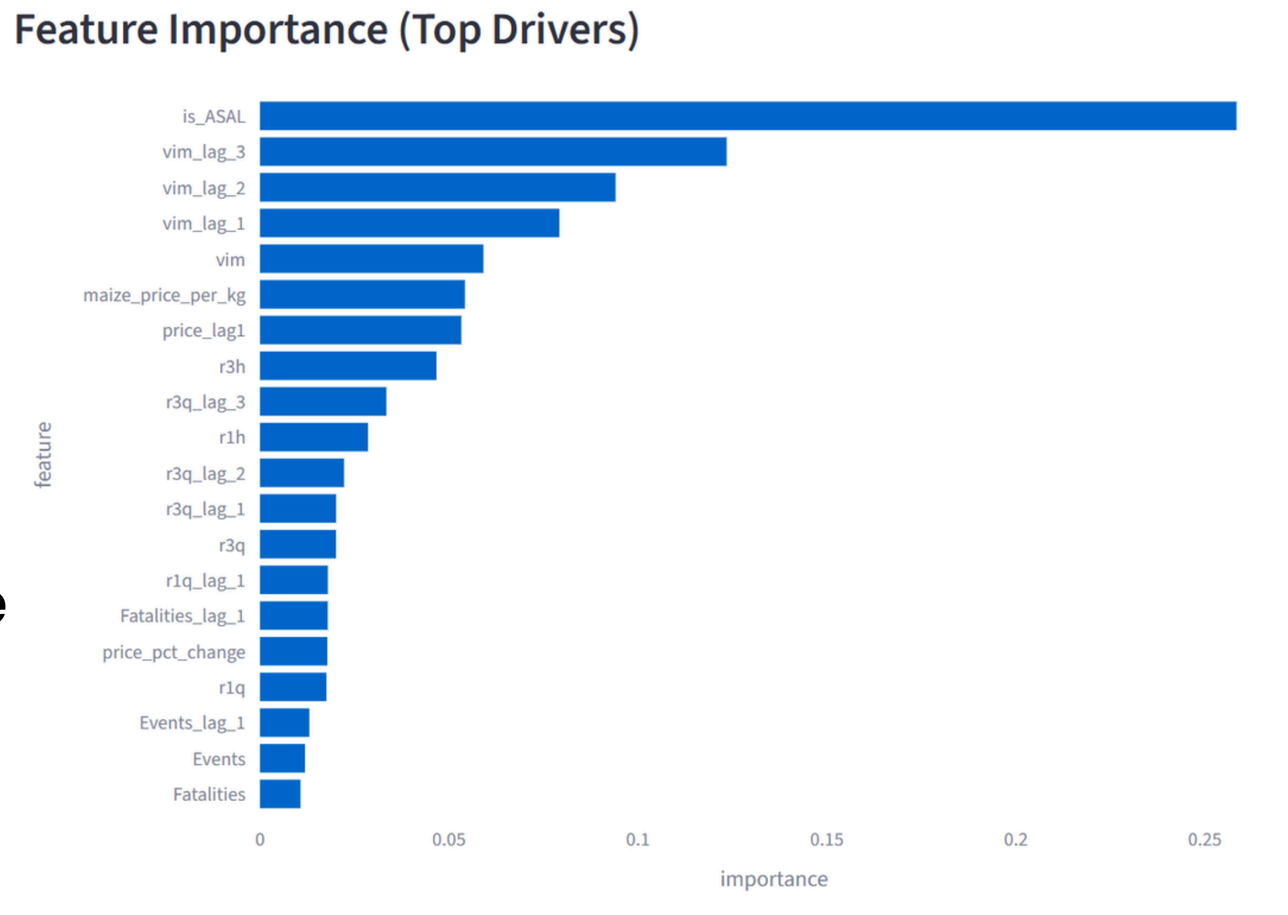
- Secure
- Watch
- High
- Extreme

METHODOLOGY

- Data Cleaning & Harmonization
- Feature Engineering (lag variables)
- Train/Test Split (time-aware validation)
- Models Compared:
- Logistic Regression
- Random Forest
- XGBoost
- Evaluation: Accuracy, Precision, Recall, ROC-AUC

KEY RESULTS

- Machine learning models improved prediction over baseline.
- Climate shocks + maize price spikes strongly associated with IPC 3+.
- Conflict contributes but less consistently than climate & prices.
- Model can flag high-risk counties one month in advance.



CONCLUSION & IMPACT

- Predictive analytics improves preparedness.
- Supports humanitarian agencies & county governments.
- Enables proactive food assistance allocation.
- Scalable to other drought-prone regions in East Africa.

THANK YOU

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